

Gurpreet Singh

AI Engineer | MLOps

gurpreetjoey@gmail.com • [LinkedIn](#) • [GitHub](#) • [Portfolio](#)

TECHNICAL SKILLS

AI/ML Engineering: LangChain, LangGraph, LlamaIndex, RAG Pipelines, MCP Servers, HuggingFace Transformers, Sentence Transformers, Prompt Engineering

MLOps & DevOps: Docker, Kubernetes (K8s), GitHub Actions CI/CD, MLflow, DVC, Prometheus, Grafana, Rolling Deployments, Auto-scaling

Backend & APIs: FastAPI, PostgreSQL, Redis, WebSockets, REST APIs, Celery, Async Python, Pydantic

ML Libraries: PyTorch, Scikit-learn, Pandas, NumPy, OpenAI API, Groq API, PEFT/LoRA Fine-tuning

Vector DBs & Cloud: ChromaDB, Pinecone · Railway, Render, AWS (S3, EC2), GCP, Vercel

Languages & Tools: Python (Advanced), SQL, Bash, Git, Linux, VS Code, JetBrains

PROJECTS

Sentiment Analysis API — Production MLOps System

FastAPI · Docker · Kubernetes · GitHub Actions · Scikit-learn · Render

- Deployed production sentiment classification API with FastAPI async endpoints and Kubernetes orchestration — class balancing techniques improved minority class recall by 22% on imbalanced dataset
- Full CI/CD pipeline with Docker containerization and K8s rolling deployments maintaining sub-200ms P95 latency

Fullstack GenAI App — AI-Powered Full Stack Application

React · Node.js · MongoDB · Express · OpenAI API · Vercel

- Built a full-stack generative AI application with React frontend, Node/Express backend, and MongoDB — integrating OpenAI API for intelligent, context-aware responses
- Deployed end-to-end on Vercel with responsive UI, REST API design, and real-time AI output streaming

github.com/gurpreet-singh-ji/Fullstack-GenAI-App

Vehicle Insurance Risk — End-to-End MLOps Pipeline

Python · Scikit-learn · Docker · Kubernetes · MLflow · DVC · GitHub Actions · FastAPI

- Built complete ML lifecycle pipeline — data ingestion, feature engineering, model training, evaluation, and automated Kubernetes deployment with rolling updates and self-healing pods for zero-downtime model releases
- Engineered GitHub Actions CI/CD pipeline with automated pytest suite, 85%+ accuracy threshold gate, Docker image build and push to registry, and kubectl rolling deployment — no broken code ever reaches production
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OPEN SOURCE CONTRIBUTIONS

LFX Mentorship — Runner Up 2024

CNCF Ecosystem (Kmesh + Linkerd) · Cloud Native Computing Foundation

- Contributed to Kmesh (eBPF-based service mesh) and Linkerd — fixed bugs in core networking logic and improved developer documentation for onboarding new contributors
- Runner-up recognition at LFX Mentorship program — competed globally among open source contributors across CNCF projects

ACHIEVEMENTS & RECOGNITION

- LFX Mentorship Runner-Up — globally recognized open source contribution to CNCF projects (Kmesh + Linkerd)
- 3 production AI/MLOps systems deployed with live URLs — K8s, MCP servers, RAG, and LLMOps as a 2nd-year student
- IBM Certified in AI — hands-on Kubernetes + MCP + LLMOps experience typically seen at senior level

EDUCATION

Indira Gandhi National Open University (IGNOU) · Expected 2028 Bachelor of Computer Applications (BCA) — India, Remote